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Mini Project

on

"Titanic Survival Prediction using Machine Learning"

-Submitted by-

Aniket B. Pathak **Roll No: CS3223**

Class and Division: TY Div B

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1. Title

Titanic Survival Prediction using Machine Learning

2. Abstract

The sinking of the RMS Titanic in 1912 caused the death of most of its passengers. Survival was not completely random: passengers of some groups, such as women and first-class travellers, had a much better chance of surviving. This mini project uses **supervised machine learning** to predict whether a Titanic passenger survived or not from details such as ticket class, gender, age, family members on board, fare and port of embarkation.

The dataset of 891 passengers was cleaned, explored with graphs and split into 80% training and 20% testing data. Three classification algorithms, Logistic Regression, Decision Tree and Random Forest, were trained and compared using accuracy, precision, recall and F1-score. **Random Forest gave the best result with 82.7% accuracy** on unseen test data. Gender, fare and age were found to be the most important factors for survival.

3. Introduction

Machine Learning (ML) is a branch of Artificial Intelligence in which a computer learns patterns from data instead of being programmed with fixed rules. **Supervised learning** is the type of ML in which the model is trained on data that already contains the correct answer (called the label or target). After training, the model can predict the answer for new, unseen data.

When the answer belongs to a fixed set of categories, for example Survived or Died, the task is called **classification**. This project is a binary classification problem: the output is either 1 (survived) or 0 (died).

The Titanic dataset is one of the most popular beginner datasets in machine learning because it is small, easy to understand and contains a mix of numerical and categorical features, missing values and a clear target column.

4. Problem Statement

To build and compare supervised machine learning classification models that predict whether a passenger on the Titanic survived or not, using passenger details such as ticket class, gender, age, number of siblings/spouses and parents/children on board, fare paid and port of embarkation.

5. Objectives

- To understand and apply supervised learning (classification) on a real-world dataset.
- To clean the data by handling missing values and converting text columns into numbers.
- To explore the data using graphs and find which factors affected survival.
- To train three classification models: Logistic Regression, Decision Tree and Random Forest.
- To compare the models using accuracy, precision, recall and F1-score and select the best one.
- To use the best model to predict survival for new passengers.

6. Data set and Libraries required

Data set

The Titanic passenger dataset (file: titanic_raw.csv) contains **891 passengers and 12 columns**. Source: https://github.com/datasciencedojo/datasets (titanic.csv).

Column	Description
Survived	Target: 0 = Died, 1 = Survived
Pclass	Ticket class: 1 (upper), 2 (middle), 3 (lower)
Name, Ticket, PassengerId	Identification details (not useful for prediction)
Sex	Gender of the passenger
Age	Age in years
SibSp	Number of siblings / spouses aboard
Parch	Number of parents / children aboard
Fare	Ticket fare paid
Cabin	Cabin number (mostly missing)
Embarked	Port of embarkation: S = Southampton, C = Cherbourg, Q = Queenstown

Libraries and tools

- **Python 3** with **Jupyter Notebook** as the working environment
- **pandas**: loading and cleaning the data
- **matplotlib** and **seaborn**: drawing graphs
- **scikit-learn**: train-test split, ML models and evaluation metrics

7. Approach/ Methodology

The project follows the standard supervised learning workflow:

Load data → Clean data → Explore (graphs) → Split into train/test → Train models → Evaluate and compare → Predict

- **Data cleaning:** missing Age values were filled with the median age, the 2 missing Embarked values with the most common port, and the Cabin column was dropped because about 77% of it was missing. PassengerId, Name and Ticket were removed as they do not help prediction.
- **Encoding:** machine learning models understand only numbers, so Sex (male = 0, female = 1) and Embarked (S = 0, C = 1, Q = 2) were converted to numeric values.
- **Exploratory data analysis:** bar charts and histograms were used to study survival by gender, class and age.
- **Train-test split:** 80% of the data (712 passengers) was used to train the models and 20% (179 passengers) was kept aside to test them on unseen data.
- **Model training:** Logistic Regression (uses a mathematical formula to separate the two classes), Decision Tree (asks a series of Yes/No questions to reach a decision) and Random Forest (many decision trees vote and the majority answer wins).

- **Evaluation:** Accuracy (correct predictions out of all), Precision (when the model says Survived, how often it is right), Recall (out of all who really survived, how many the model found) and F1-score (balance of precision and recall).

8. Implementation

1. Loading and cleaning the data

```
import pandas as pd
df = pd.read_csv("titanic_raw.csv")

df["Age"] = df["Age"].fillna(df["Age"].median())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
df = df.drop(columns=["Cabin"])

df["Sex"] = df["Sex"].map({"male": 0, "female": 1})
df["Embarked"] = df["Embarked"].map({"S": 0, "C": 1, "Q": 2})
```

Column	Missing values	Action taken
Age	177	Filled with median age
Embarked	2	Filled with most common port
Cabin	687	Column dropped

2. Exploratory data analysis

Only about 38% of the passengers survived and 62% died. Women had a survival rate of about 74% compared with about 19% for men. Survival also fell with ticket class: about 63% in first class, 47% in second class and 24% in third class.

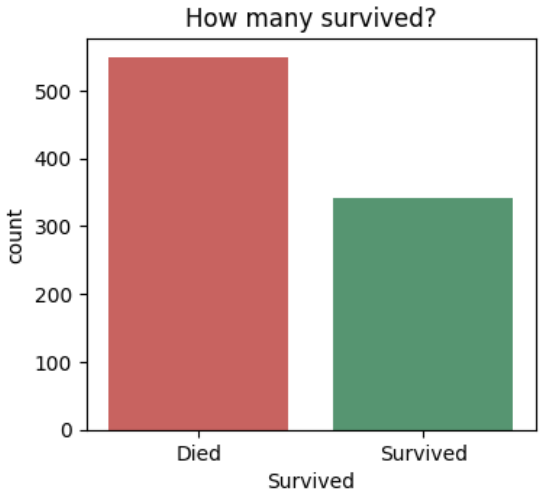


Figure 1: Survival count (0 = Died, 1 = Survived)

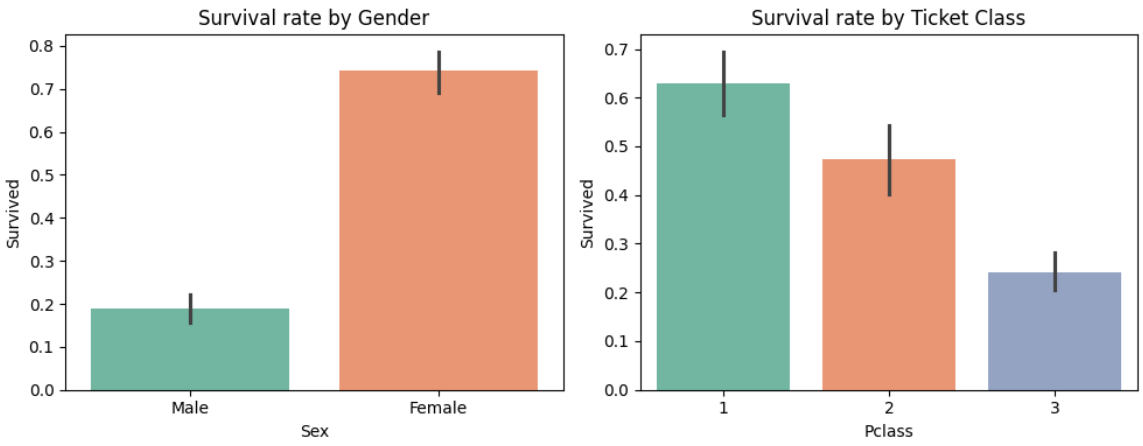


Figure 2: Survival rate by gender and by ticket class

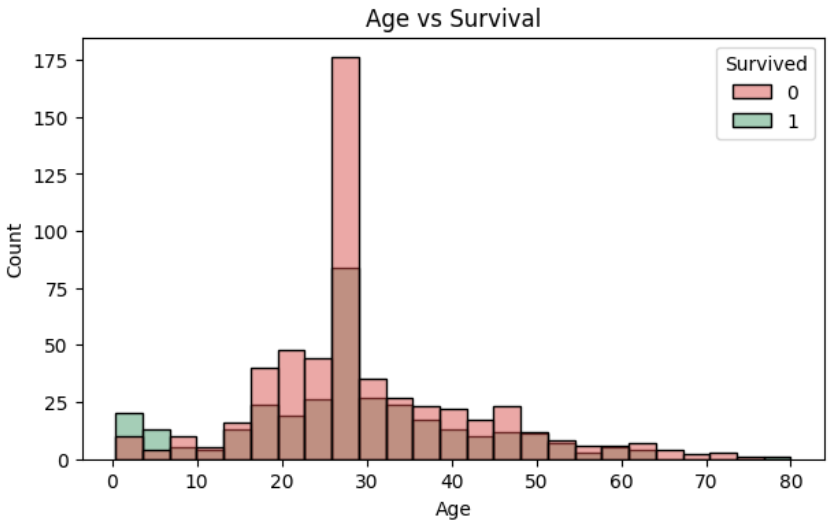


Figure 3: Age distribution of passengers who survived and died

3. Train-test split

```
from sklearn.model_selection import train_test_split

X = df[["Pclass", "Sex", "Age", "SibSp", "Parch", "Fare", "Embarked"]]
y = df["Survived"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

4. Training and evaluating the models

```
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score
from sklearn.metrics import recall_score, f1_score

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Decision Tree": DecisionTreeClassifier(max_depth=4, random_state=42),
    "Random Forest": RandomForestClassifier(n_estimators=100, random_state=42),
}
for name, model in models.items():
    model.fit(X_train, y_train)          # learn from training data
    pred = model.predict(X_test)         # predict on unseen test data
    print(name, accuracy_score(y_test, pred))
```

9. Results

All three models were tested on the same 179 unseen passengers.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.8268	0.8116	0.7568	0.7832
Logistic Regression	0.7989	0.7714	0.7297	0.7500
Decision Tree	0.7989	0.8393	0.6351	0.7231

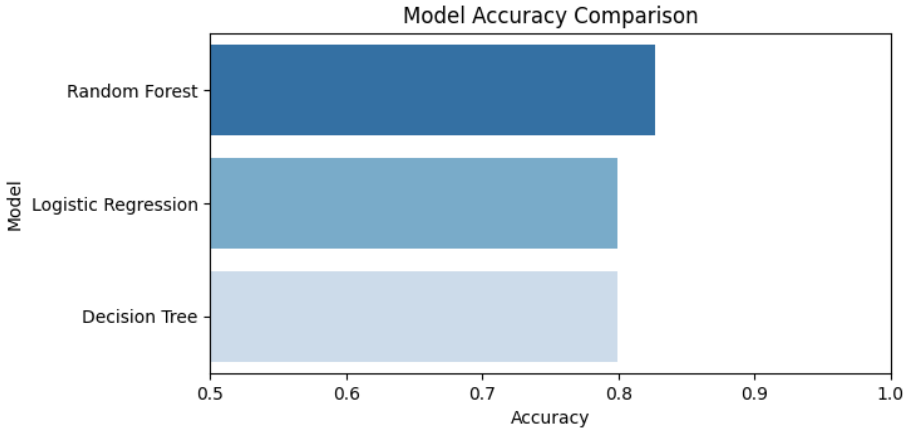


Figure 4: Accuracy comparison of the three models

Random Forest is the best model with **82.7% accuracy**: 148 of the 179 test passengers were predicted correctly. Logistic Regression and Decision Tree both reached 79.9%.

Confusion matrix (Random Forest)

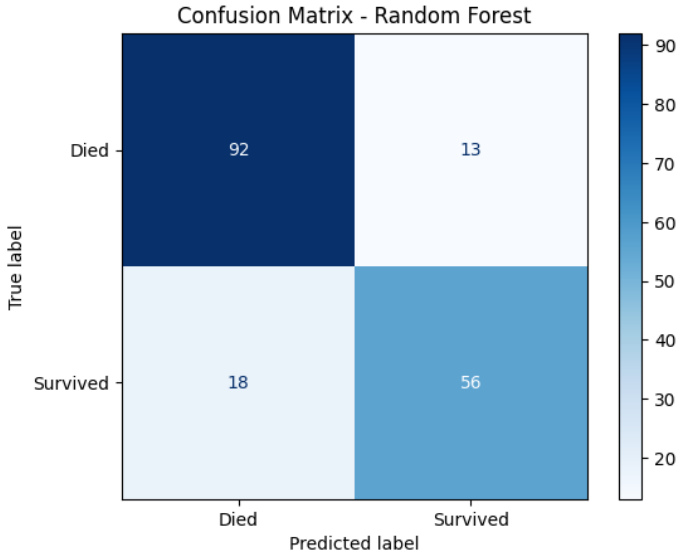


Figure 5: Confusion matrix of the Random Forest model

Of the 179 test passengers, 92 who died and 56 who survived were predicted correctly (148 correct). The model wrongly predicted 13 passengers as survived when they died, and 18 as died when they actually survived.

Decision tree

The decision tree is easy to understand because it shows the questions the model asks, such as whether the passenger is female or male, the ticket class and the age. It is shown below with depth 3.

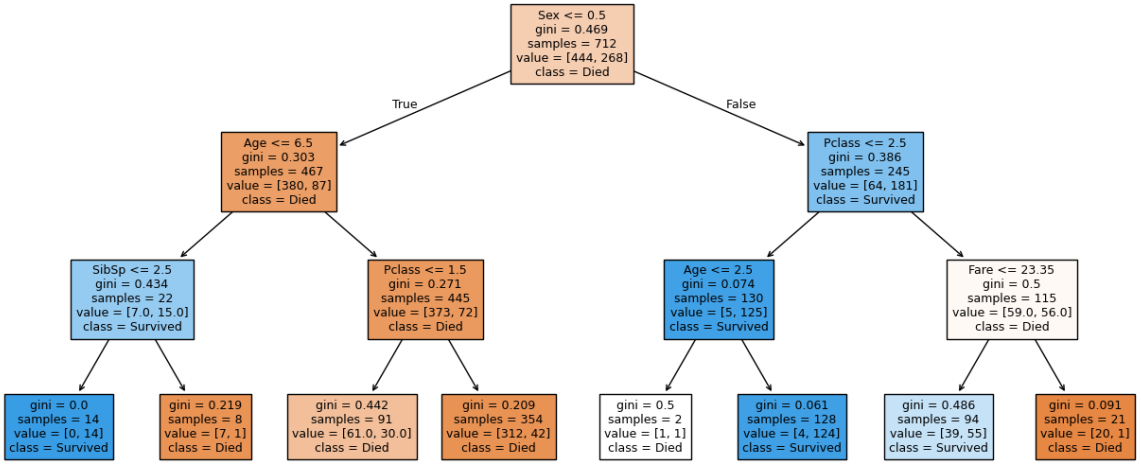


Figure 6: Decision tree (depth 3)

Important features

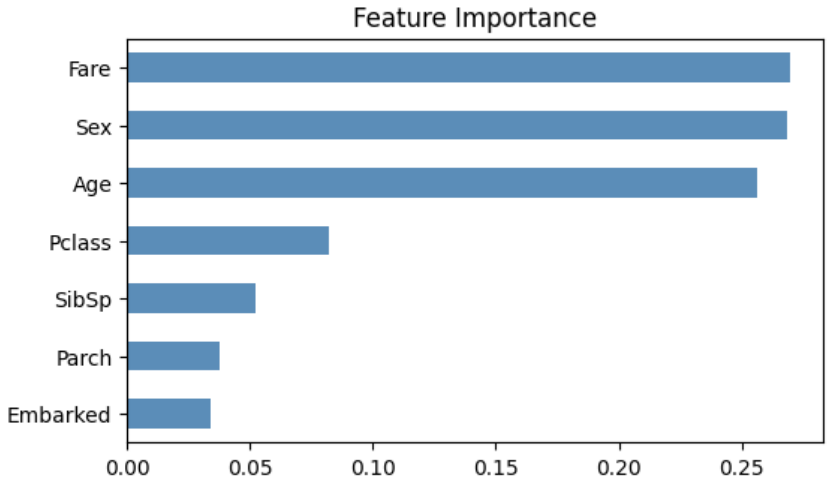


Figure 7: Feature importance from the Random Forest model

Fare (0.27), Sex (0.27) and Age (0.26) are the three most important features, followed by Pclass (0.08). In the graphs, gender showed the strongest effect on survival, which matches the "women and children first" rule followed during the evacuation.

Prediction for new passengers

Passenger	Predicted result
1st class woman, age 28	Survived
3rd class man, age 30	Died
2nd class girl, age 8	Survived

10. Conclusion

In this project, supervised machine learning (classification) was used to predict Titanic passenger survival. After cleaning the data and exploring it, three models were trained and tested on unseen passengers. **Random Forest performed best with 82.7% accuracy**, ahead of Logistic Regression and Decision Tree (79.9% each).

The analysis showed that gender, fare and age are the most important factors, and that women, children and first-class passengers had a much better chance of survival.

Limitations: the dataset is small (891 passengers) and the models were used with basic settings. **Future scope:** trying more algorithms such as KNN and SVM, tuning model parameters and creating new features such as family size and passenger title.